INCH-POUND

MIL-P-20442B (AR)
AMENDMENT 4
31 July 1990
SUPERSEDING
AMENDMENT 3
30 December 1988

MILITARY SPECIFICATION

PLUGS, LIFTING (EYEBOLT TYPE) FOR PROJECTILES

This Amendment forms a part of Military Specification MIL-P-20442B (MU), dated 8 May 1968, and is approved for use within the US Army Armament Munitions and Chemical Command, and is available for use by all Departments and Agencies of the Department of Defense.

PAGE 1

2.1 Specifications, add the following:

"FEDERAL

*	QQ-S-781	- Strapping, Steel, and Seals
	PPP-B-587	- Boxes, Wood, Wirebound Pallet Type
	TT-C-490	- Cleaning Methods and Pretreatment of Ferrous Surface for Organic Coatings

MILITARY

	MIL-B-121	- Barrier Material, Water, Vaporproof Flexible
	MIL-B-13239	- Barrier Material, Waterproofed, Flexible all Temperature
*	MIL-A-48078	- Ammunition, Standard Quality Assurance Provisions, General Specification For"

Delete: "MIL-I-45607-Inspection Equipment, Supply and Maintenance of"

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Drawings, Army, 10520911: Delete "Detonator" and substitute "Detector".

Publications, Army:

Delete: "ET-6112851 - Equipment Tabulation" in its entirety.

Add paragraphs 2.2 and 2.3 as follows:

"2.2 Other publications. The following document(s) form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted shall be those listed in the issue of the DODISS specified in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS shall be the issue of the nongovernment documents which is current on the date of the solicitation.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM B-117 - Salt Spray (Fog) Testing

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein (except for associated detail specifications, specification sheets or MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained."

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4.1 and 4.1.1, Delete in their entirety and substitute the following:

"4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The

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Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material."

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4.2.1, lines 4 and 5: Delete "assemblies and twenty-five each of every component" and substitute "lifting plugs (20 plugs unphosphated and unpainted and 5 phosphated and painted)."

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4.2.3 Delete in its entirety and substitute the following:

"4.2.3 Rejection. See MIL-A-48078."

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4.3.2.1, 4.3.2.2, 4.3.2.3, 4.3.2.4, 4.3.2.5, 4.3.2.6 and 4.3.2.7. Delete the last column "Code No." in its entirety.

4.3.2.1:

Major Category: Add "107 Perpendicularity of pitch diameter of threads with shoulder...Gage. 108 Distance from thread undercut to outside of flange 360 min...Gage."

Minor Category: Add "209 Chamfer missing...Visual."

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4.3.2.4, Minor Category:

201: Delete "and collar"
206: Delete "Gage" and substitute "Visual".

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4.3.2.5, Major Category: Delete Major Defect 108 in its entirety and substitute the following:

"108 Perpendicularity of pitch diameter to collar...Gage."

Add Major Defect 111 to read as follows:

"111 Perpendicularity of pitch diameter to shoulder...Gage."

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Add paragraphs 4.3.3.2 and 4.3.3.2.1 as follows:

"4.3.3.2 Protective coating - salt spray. The sample size and acceptance criteria shall be as specified in the applicable specification. If failure occurs, all items processed since last acceptance shall be rejected and corrective action taken.

4.3.3.2.1 Salt spray. After qualification in accordance with TT-C-490, two (2) salt spray samples shall be taken each week, one during the first 30 minutes of production and a second at any other time. Failure to comply with the requirements of the applicable drawing shall be cause for taking the corrective action prescribed in TT-C-490."

4.3.4, 4.3.4.1 and 4.3.4.2, Delete in their entirety and substitute the following:

"4.3.4 Inspection equipment. The inspection equipment required to perform the examinations and tests prescribed herein is described in the "Method of Inspection" column in the table of paragraph 4.3.2.1. The contractor shall submit for approval inspection equipment designs in accordance with the terms of the contract. See Section 6 of MIL-A-48078 and 6. - herein."

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Add paragraph 4.4.4 as follows:

"4.4.4 Salt spray testing. Salt spray testing shall be accomplished on production plugs that were acceptable at the time of painting. The salt spray test shall be performed utilizing the equipment and procedures specified by ASTM method B-117 and TT-C-490 with the exception that test samples shall be selected in accordance with 4.3.3.2.1. The number of hours and the percentage of sodium chloride solution shall be as indicated on the applicable drawing. Examination shall be made for any evidence of corrosion of the painted areas."

Delete Section 5 in its entirety and substitute the following:

"5. PACKAGING

5.1 Preservation. Preservation shall be Level C.

5.1.1 Packaging for lifting plugs. Lifting plugs shall be packaged and packed in accordance with Standard MIL-STD-1169.

5.1.2 Alternate packaging for lifting plugs. Alternatively, Type G lifting plugs shall be packed in a wood, pallet type, wirebound box with interior packing as described herein.

5.1.2.1 Trays. 36 lifting plugs shall be placed in a tray, Panta-Pak Number L. A. 36 cavity, make from 10-mil thick natural color polyethylene 13 1/2 inches by 16 1/8 inches (see 6.4); trays of equivalent design may also be used. The tray shall be placed in a 125-pound test, B flute, corrugated fiberboard tray. The tray shall be made from a corner cut sheet 17 5/8 by 20 1/4 inches, outside dimensions, cut and scored to form 2-inch flaps. The inside tray dimensions when the flaps are folded 90 shall be 16 1/4 by 13 5/8 inches.

5.1.2.2 Shipping container. 42 trays (1512 lifting plugs), shall be arranged 2 by 3 trays in a layer, 7 layers high, in a wirebound pallet box conforming to the general requirements of Type 1, Class 2 of Federal Specification PPP-B-587 and the following requirements. The inside dimensions shall be approximately 32 3/4 by 41 1/4 by 23 1/8 inches. All dimensions for the base, sides and top stated below are minimum dimensions. A sheet of CW 2 or T2 barrier paper complying with requirements of Military Specification

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MIL-B-13239 shall be located across the top of the load and completely cover the top layer of trays. Alternately, Type 2, Grade A, Class 1 barrier material complying with Military Specification MIL-B-121 may be used. Box assembly shall include driving a minimum of three, 2 1/2 inch long, 11 gage (.120 diameter) drive screw nails through the sheathing and bottom cleat of each side section into the adjacent riser batten of the base. Two 3/4 x .028 steel straps complying with Type I of IV, Class A or C, of Federal Specification QQ-S-781 shall be applied parallel with the stringers, around the box over the 1 1/8 x 7/8 inch battens of the top.

5.1.2.2.1 Pallet base. The pallet base shall be constructed with three stringers 3 5/8 by 1 5/8. Three riser battens 2 3/4 wide by 15/16 thick shall be located at 90 to the stringers and 17/32 deck material located at 90 to the riser battens Bottom deck boards are not required.

5.1.2.2.2 Pallet sidewalls. The four-section box sidewalls shall consist of two edge and two intermediate 13/16 by 7/8 inch cleats at each section. The cleat dimensions from the bottom cleat to the intermediate cleats shall be approximately 6 and 16 inches. Sheathing shall consist of eight 3 5/8 wide by 3/8 inch thick boards at the corners of the made-up box. Additionally, two boards of like dimensions shall be located about 8 inches in from each outside corner of the longest sections of the box (front and back). The remaining portions of each box panel shall be sheathed with 7/32 veneer. One 12 gage galvanized binding wire ending in wide and narrow loops shall be stapled over row of cleats, and one at the approximate center of the panel. Staple spacing shall not exceed 1 1/2 inches.

5.1.2.2.3 Pallet top. The top shall be made of sheathing material which is 5/16 inch resawn boards stapled to three 2 1/4 wide x 3/8 inch thick liners equally spaced and two 1 1/8 inch x 7/8 inch thick battens, with ends mitered at 45 degrees, located midway between the outside and intermediate 3/8 inch thick liners. The liners and battens shall be on the outside of the top with the sheathing fitted inside the side cleats. Liners shall extend over the top cleats of the front, back and sides.

5.1.3 Marking. Marking shall be in accordance with Standard MIL-STD-1169.

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6.1, 6.2, 6.3, 6.3.1, 6.3.2, 6.3.3, 6.3.4 and 6.3.5, Delete in their entirety and substitute the following:

"6.1 Ordering data. See MIL-A-48078 (AR).

6.1.1 Data card. Data cards are required and shall be prepared. See MIL-A-48078 (AR).

6.2 Submission of inspection procedures and equipment designs for approval. See MIL-A-48078. Submit equipment designs, as required, to Commander, U.S. Army Armament, Research, Development and Engineering Center, ATTN: AMSMC-QAR-I(D), Picatinny Arsenal, NJ 07806-5000. This address will be specified on the Contract Data Requirements List, DD Form 1423 in the contract. Unless otherwise specified, data item DI-R-1714 will apply."

Section 6, add:

"6.4 Panta-Pak trays. Panta-Pak trays are a product of the Pantacote Company, Panta-Pak Division, 26 Jefferson Street, Passaic, New Jersey 07056 (see 5.1.2.1)."

The margins of this amendment are marked with an asterisk or vertical line to indicate where changes (additions, modifications, corrections, deletions) from the previous amendment were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous amendment.

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